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RESEARCH ARTICLE

Beyond Precision and Recall: Measuring Search Engine Consistency Using Rank Stability

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ABSTRACT Traditional information retrieval metrics assess document relevance but neglect ranking stability—a critical factor in today's dynamic web environments where search algorithms constantly evolve. We present *Rmeasure*, a comprehensive framework that quantifies search engine consistency by analyzing both overlapping and non-overlapping results through the lens of psychophysical principles, particularly the Weber-Fechner Law. Our three-component approach (*Roverlap*, *Rnon-overlap*, and *Rcomprehensive*) provides multidimensional insights into ranking variations. Experimental evaluation across Google and Bing using diverse query types reveals significant performance differences: Bing demonstrates superior ranking stability (minimum *Roverlap* of 0.0527), while Google exhibits considerable fluctuation (*Rnon-overlap* reaching 0.3653), especially for high-volume queries. Statistical measures confirm Bing's consistency advantage, with Google showing rank differences averaging up to 21.622 positions. Beyond technical contributions, *Rmeasure* advances UN Sustainable Development Goals by enhancing information access equity (SDG 4), fostering innovation in search technology evaluation (SDG 9), and promoting consistent retrieval experiences across diverse user populations (SDG 10). By integrating perceptual modeling with rank-based analytics, *Rmeasure* extends search engine evaluation beyond traditional relevance metrics, offering a robust methodology for assessing real-world search performance.

INDEX TERMS Search engine evaluation, rank correlation, rank stability, repeated queries, ranking consistency, overlapping results, non-overlapping results, psychophysical modeling, Weber-Fechner Law, information retrieval, Rmeasure framework, Google, Bing.

I. INTRODUCTION

Search engine ranking inconsistency poses a significant challenge in the modern information ecosystem, undermining user experience and information retrieval effectiveness despite the revolutionary impact of the World Wide Web (WWW) on information access [1]. While multiple search engines compete for market share, users typically develop loyalty to a specific platform based on perceived reliability and result quality. However, this perceived reliability is continuously tested by dynamic ranking algorithms, real-time indexing updates, and personalization mechanisms that produce inconsistent results across identical queries. Evaluating and quantifying this ranking stability represents a critical yet underexplored dimension of search engine performance assessment.

Traditional Information Retrieval (IR) metrics, such as precision, recall, and F-measure, focus primarily on document relevance but fail to capture the temporal stability of search rankings [2]. These conventional metrics rely on predefined relevance judgments that are inherently subjective and context-dependent. Furthermore, they inadequately account for the ranked order of retrieved results—a factor that significantly influences user behavior, as research demonstrates

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that users predominantly interact with the top-ranked search results, often ignoring lower-ranked alternatives [3]. As a result, these established IR metrics provide an incomplete picture of search engine performance from a user-centric perspective.

The variability in search results across repeated identical queries represents a fundamental limitation of current search technologies. When users submit the same query multiple times, they may encounter different rankings or entirely different result sets due to continuous indexing updates, personalization algorithms, and real-time ranking adjustments. These frequent fluctuations can erode user confidence and create frustration when attempting to relocate previously discovered information. Despite its significant impact on user experience and information accessibility, the dimension of search result stability remains inadequately addressed in contemporary evaluation frameworks. This research gap necessitates the development of robust metrics that can effectively quantify rank stability across temporally separated searches.

Existing rank correlation measures, including Kendall's Tau and Spearman's Rho, offer mathematical frameworks for quantifying similarity between ranked lists [4]. However, these traditional measures operate under the assumption that both compared lists contain identical elements arranged in different orders—an assumption that rarely holds in web search scenarios. Modern search engines continuously index an expanding universe of documents, resulting in both overlapping and non-overlapping results across search instances [5]. Consequently, applying conventional rank correlation measures without appropriate modification can yield incomplete or potentially misleading insights into search result consistency.

The phenomenon of ranking inconsistency can be understood through the lens of psychophysical principles, particularly the Weber-Fechner Law, which establishes that perceived intensity follows a logarithmic function of actual stimulus intensity [6]. When applied to search ranking behavior, this principle suggests that initial query repetitions produce noticeable ranking variations, but subsequent repetitions yield progressively smaller perceptible changes. This pattern indicates that search engines dynamically adjust their rankings over repeated queries before eventually stabilizing. Incorporating this psychological insight into search evaluation methodologies enables a more nuanced and realistic assessment of ranking stability across time.

Recent empirical studies on search engine ranking dynamics further highlight the need for enhanced evaluation metrics [7]. For instance, Dean's [8] comprehensive analysis of 11.8 million Google search results identified content quality, backlink profiles, and user engagement signals as key ranking factors that contribute to result variability. These findings align with our research focus on ranking stability and validate the need for specialized metrics. Additionally, advanced rank correlation coefficients such as AP correlation (τ_{AP}) have emerged, offering probabilistic interpretations of ranking consistency that better reflect real-world user interactions [9]. These developments underscore the evolving nature of search evaluation methodologies and the critical need for novel measures that account for the complex, dynamic behaviors of modern search algorithms.

To address these identified gaps, this study introduces a novel rank correlation measure specifically designed for evaluating web search engines. Unlike conventional correlation measures, our approach separately accounts for both overlapping and unique documents while incorporating the effects of query repetition over time. By capturing the systematic rank shifts introduced by search engines in response to repeated identical queries, our method provides a more comprehensive assessment of ranking stability. We validate our approach through empirical experiments on major search engines including Google and Bing, offering new insights into search consistency patterns across platforms.

The key contributions of this research are as follows:

1) Identification and analysis of fundamental limitations in conventional IR metrics for evaluating search engine performance, with particular emphasis on the need for rank stability assessment.
2) Integration of Weber-Fechner Law principles to explain and model ranking fluctuations resulting from query repetition effects.
3) Development of a novel rank correlation measure that independently evaluates both overlapping and non-overlapping search results to provide a more granular assessment of ranking stability.
4) Comprehensive empirical evaluation of ranking consistency across major commercial search engines, revealing platform-specific patterns and providing actionable insights for search engine optimization and user experience enhancement.

The remainder of this paper is structured as follows: Section II reviews related work on search engine evaluation and rank correlation measures. Section III presents the proposed rank correlation approach and its mathematical formulation. Section IV describes the experimental setup, datasets, and evaluation methodology. Section V discusses the results and key findings. Finally, Section VI concludes the paper and suggests potential directions for future research.

II. RELATED WORK

The evaluation of search engine performance has traditionally centered around relevance-based metrics such as precision, recall, and F-measure. These metrics assess retrieval accuracy effectively but fail to capture the temporal or perceptual stability of ranking, a critical dimension of user experience in dynamic search environments. As search engines continuously evolve through real-time indexing, algorithmic experimentation, and personalization, ranking consistency needs more sophisticated evaluation frameworks.

Correlation coefficients are widely used in statistical analysis to measure relationships between variables. Pearson's correlation coefficient (r) is suitable for linear relationships between continuous variables [10]. For ranking similarity, Kendall's Tau and Spearman's Rho are popular non-parametric alternatives [9]. However, both assume full overlap between ranked lists—an assumption rarely satisfied in web search due to ongoing index changes and query contextualization.

To mitigate these limitations, variations such as Weighted Kendall's Tau [11] and adaptive correlation measures [12] were introduced, prioritizing top-ranked items and adapting to shifting rankings. Yet, they still underperform in settings with significant non-overlap between result sets. Moreover, their distance functions do not incorporate perceptual models of user sensitivity to changes in rank.

Psychophysical principles, such as the Weber-Fechner Law [13], have been used to explain how perceived intensity varies logarithmically with stimulus changes. This model suggests that users are more sensitive to ranking differences at the top of a result list. Incorporating such principles into ranking evaluation—e.g., through attention-weighted or perception-aware metrics—offers a more user-aligned assessment framework [14].

Beyond traditional correlation metrics, more robust methods have emerged. *AP Correlation* (τ_{AP}) accounts for the average precision between ranked lists and reflects early-position sensitivity [15]. *Rank-Biased Overlap (RBO)* supports incomplete and indefinite lists using a probabilistic decay model and has gained popularity for handling non-overlapping ranks effectively [16]. Semantic similarity-based techniques [17] and probabilistic inference models [18] further enhance alignment across evolving results but typically require external knowledge bases or complex models.

Table 1 summarizes key characteristics of major rank correlation techniques.

Scalable frameworks have also been explored. Adaptive sampling methods extract representative subsets for evaluation [19], while progressive evaluation frameworks assess search rankings as results arrive incrementally [20]. Such techniques reduce computational cost while maintaining evaluation robustness in expanding web corpora.

Modern research trends emphasize automation and context adaptation in search evaluation. Emerging methods include machine learning-based metric selection [21], real-time evaluation tools [22], and context-aware assessment [23], aligning ranking behavior with personalized user expectations.

Despite these advances, no existing metric unifies:

- Rank-based stability for both overlapping and non-overlapping results,
- Temporal sensitivity across repeated queries,
- Perception-aware modeling of rank changes.

To address these gaps, we introduce **Rmeasure**, a novel rank correlation framework comprising *Roverlap*, *Rnon-overlap*, and *Rcomprehensive* components. Rmeasure uniquely integrates statistical variation and psychophysical modeling to evaluate the consistency of web search rankings across repeated queries, even when the result sets are only partially overlapping.

III. MATHEMATICAL PROOF OF TRADITIONAL IR METRICS' INADEQUACY USING INFORMATION GAIN/LOSS

Traditional Information Retrieval (IR) metrics such as precision, recall, and F1-score evaluate document relevance but fail to assess ranking stability. Users expect a reasonable level of consistency in search results over repeated queries. However, search engines dynamically update their rankings, leading to variations that conventional metrics fail to capture. This section establishes a formal mathematical proof demonstrating that traditional IR metrics remain unchanged despite search result instability.

A. INFORMATION TURNOVER FRAMEWORK

To measure ranking stability, we introduce an **Information Turnover Framework**, which quantifies document changes in retrieved results across repeated queries.

Definition 1 (Retrieved Result Set): Let Q be a query executed at two time points t_1 and t_2, yielding result sets:

$$R(Q, t_1), \quad R(Q, t_2) \tag{1}$$

where $R(Q, t)$ denotes the top n ranked documents retrieved at time t.

Definition 2 (Information Gain and Loss): We define:

$$\text{Gain}(Q, t_1, t_2) = |R(Q, t_2) \setminus R(Q, t_1)|, \tag{2}$$
$$\text{Loss}(Q, t_1, t_2) = |R(Q, t_1) \setminus R(Q, t_2)|. \tag{3}$$

where: - *Information Gain* refers to newly introduced documents at t_2 that were absent at t_1. - *Information Loss* refers to documents removed from t_1 that no longer appear in t_2.

Definition 3 (Information Turnover): Total instability in search results is computed as:

$$IT(Q, t_1, t_2) = \text{Gain}(Q, t_1, t_2) + \text{Loss}(Q, t_1, t_2). \tag{4}$$

Definition 4 (Normalized Information Turnover): To facilitate comparisons across different queries, we define a normalized version:

$$NIT(Q, t_1, t_2) = \frac{IT(Q, t_1, t_2)}{|R(Q, t_1) \cup R(Q, t_2)|}. \tag{5}$$

where $0 \leq NIT \leq 1$. A value of: - $NIT = 0$ indicates perfect stability (identical result sets across queries). - $NIT = 1$ represents complete turnover (entirely different result sets).

B. THEOREM: FUNDAMENTAL LIMITATION OF TRADITIONAL IR METRICS

Theorem III-B (Invariance of Traditional Metrics to Result Turnover): For any query Q and time points t_1 and t_2, there exist distinct result sets $R(Q, t_1)$ and $R(Q, t_2)$ such that all traditional IR metrics remain unchanged, while $IT(Q, t_1, t_2)$ can be arbitrarily large.

TABLE 1. Comparison of rank correlation metrics.

Metric	Strengths	Limitations	Overlap Required?
Kendall's Tau	Sensitive to pairwise swaps	Assumes identical elements in lists	Yes
Spearman's Rho	Considers full rank order	Insensitive to top-position changes	Yes
AP Correlation (τ_{AP})	Top-weighted, user-aligned	Partially handles missing elements	Partial
Rank-Biased Overlap (RBO)	Incomplete lists supported, top-weighted	Parameter tuning needed, no perceptual modeling	No
Semantic similarity metrics	Capture content-level equivalence	Require NLP tools, not rank-sensitive	No
Rmeasure (Proposed)	Models overlap, non-overlap, and perceptual rank change	Requires repeated queries	No

Proof: Let $k > 0$ be the number of top results considered. Assume:

$$R(Q, t_1) = \{d_1, d_2, \ldots, d_k\}, \tag{6}$$

$$R(Q, t_2) = \{d_{k+1}, d_{k+2}, \ldots, d_{2k}\}. \tag{7}$$

where all documents are relevant.

Since all retrieved documents remain relevant, the precision, recall, and F1-score are:

$$P(Q, t_1) = P(Q, t_2) = 1, \tag{8}$$

$$Re(Q, t_1) = Re(Q, t_2) = \frac{k}{|D_{\text{rel}}|}, \tag{9}$$

$$F1(Q, t_1) = F1(Q, t_2) = \frac{2k}{|D_{\text{rel}}| + k}. \tag{10}$$

However, the **Information Turnover** is:

$$IT(Q, t_1, t_2) = |R(Q, t_1) \setminus R(Q, t_2)| + |R(Q, t_2) \setminus R(Q, t_1)| = k + k = 2k. \tag{11}$$

Thus, the Normalized Information Turnover is:

$$NIT(Q, t_1, t_2) = \frac{2k}{2k} = 1. \tag{12}$$

This proves that traditional IR metrics remain unchanged despite complete instability, contradicting their effectiveness in measuring ranking stability.

C. COROLLARY: EXISTENCE OF CONTROLLED STABILITY LEVELS

Corollary III-C (Controllable Result Stability): For any stability level $0 \leq v \leq 1$, there exist result sets with identical IR metrics that exhibit a normalized information turnover exactly equal to v.

Proof: Given a result set size k, let the overlap size be:

$$|O| = |R(Q, t_1) \cap R(Q, t_2)|. \tag{13}$$

We define:

$$IT(Q, t_1, t_2) = 2(k - |O|), \tag{14}$$

$$|R(Q, t_1) \cup R(Q, t_2)| = 2k - |O|. \tag{15}$$

Substituting into (5):

$$v = \frac{2(k - |O|)}{2k - |O|}. \tag{16}$$

Solving for $|O|$:

$$|O| = \frac{2k(1 - v)}{2 - v}. \tag{17}$$

The number of changed documents is:

$$|D_{\text{change}}| = k - |O| = k - \frac{2k(1 - v)}{2 - v}. \tag{18}$$

Approximating to an integer:

$$|D_{\text{change}}| = \left\lceil \frac{vk}{2 - v} \right\rceil. \tag{19}$$

By adjusting $|D_{\text{change}}|$, we achieve any desired stability level while keeping IR metrics unchanged.

IV. R_{MEASURE}: A POSITION-BASED PERFORMANCE METRIC FOR WEB SEARCH ENGINES

Web search evaluation faces a fundamental challenge when comparing different search engines: how do we assess ranking quality when search results don't completely overlap? Traditional metrics work well when result sets are identical, but fall short when search engines return partially or entirely different web pages. R_{measure} addresses this critical gap by providing a comprehensive framework for evaluating search engine performance across both overlapping and non-overlapping results.

A. CORE PROBLEM IN SEARCH ENGINE EVALUATION

When two search engines return different sets of results for the same query, conventional metrics become insufficient because:

- They typically assume result sets are identical or highly similar
- They cannot meaningfully compare ranking quality across different document sets
- They fail to capture the psychological impact of ranking variations on users

R_{measure} resolves these limitations through a two-component approach that separately evaluates overlapping and non-overlapping results, followed by a comprehensive metric that combines these components.

B. THE R_{OVERLAP} COMPONENT

The R_{overlap} metric quantifies ranking consistency for web pages that appear in both result sets. It measures how similarly the common documents are positioned across different search engines.

$$R_{\text{overlap}} = \left[\frac{\sum_{x \in A \cap B} \frac{|r_A(x) - r_B(x)|}{n-1}}{|A \cap B|} \right] \times \frac{n}{|A \cup B|} \tag{20}$$

where:

- A and B represent the result sets from two different search engines
- $r_A(x)$ and $r_B(x)$ denote the rank positions of web page x in sets A and B
- n is the maximum number of results considered (e.g., top-10 or top-20)
- $|A \cap B|$ is the number of common results between sets A and B
- $|A \cup B|$ is the total number of unique results across both sets

The first bracketed term calculates the average normalized rank difference for overlapping results. The second term ($\frac{n}{|A \cup B|}$) applies a scaling factor that accounts for the degree of overlap between result sets.

The R_{overlap} metric produces values in the range [0, 1], where:

- $R_{\text{overlap}} = 0.0$ indicates perfect ranking consistency (all common results maintain identical rank positions)
- $R_{\text{overlap}} = 1.0$ represents maximum ranking inconsistency (common results appear at opposite ends of the rankings)

Lower R_{overlap} values signify greater ranking stability across search engines.

C. PSYCHOLOGICAL FOUNDATION: THE WEBER-FECHNER LAW

The R_{overlap} metric is grounded in the Weber-Fechner Law from psychophysics, which models how humans perceive changes in stimulus intensity:

$$S = k \log \left(\frac{I}{I_0} \right) \tag{21}$$

where:

- S represents the perceived sensation intensity
- I is the actual stimulus intensity
- I_0 is the reference intensity level
- k is a scaling constant

This logarithmic relationship explains why R_{overlap} doesn't simply use linear rank differences. Human perception of ranking changes follows a similar logarithmic pattern—small changes in top positions (e.g., from rank 1 to rank 2) are perceived as more significant than the same absolute change further down the list (e.g., from rank 9 to rank 10).

D. ACCOMMODATING QUERY REPETITION

In an ideal evaluation scenario, we would measure search result stability over extended time periods (e.g., days or weeks) to capture how ranking algorithms evolve and affect user experience. However, waiting for such long intervals is impractical for timely evaluation.

1) PRACTICAL APPROACH TO TEMPORAL STABILITY ASSESSMENT

To overcome this limitation, R_{measure} employs a pragmatic approach: instead of waiting for extended periods, we repeatedly submit identical queries within a short timeframe. This technique serves as a proxy for temporal stability assessment, as modern search engines typically exhibit variations in results even for identical queries executed in quick succession. These variations arise from:

- Index updates that occur continuously rather than at fixed intervals
- Load balancing across distributed server farms
- Randomized elements in ranking algorithms
- A/B testing of algorithm modifications
- Personalization factors that may fluctuate even within short periods

By analyzing these short-term variations, we can infer how the search engine might behave over longer time horizons without waiting for those periods to elapse. This approach provides a practical approximation of temporal stability while enabling timely evaluation.

2) STATISTICAL REFINEMENTS FOR REPEATED QUERIES

To quantify performance across repeated queries, R_{measure} introduces several statistical refinements:

$$RO_{\text{minmin}} = \min(RO) \tag{22}$$

$$RO_{\text{minmax}} = \max(RO) \tag{23}$$

$$RO_{\text{ave}} = \frac{1}{|RO|} \sum_{Y \in RO} Y \tag{24}$$

where RO represents the set of all R_{overlap} values calculated across multiple query repetitions. The size of this set is determined by:

$$|RO| = \binom{rf}{2} = \frac{rf(rf-1)}{2} \tag{25}$$

With rf representing the repetition factor (how many times each query is executed).

3) INTERPRETATION OF QUERY REPETITION METRICS

These statistical measures provide valuable insights into search engine stability:

- RO_{minmin}: The best-case scenario for ranking consistency
- RO_{minmax}: The worst-case scenario for ranking consistency
- RO_{ave}: The average ranking consistency across all query repetitions
- RO-Diff: The spread between these values, indicating overall stability

A stable search engine will show minimal variation in R_{overlap} values across query repetitions, while an unstable one will exhibit significant fluctuations. These short-term fluctuations serve as a reasonable proxy for long-term stability, allowing evaluators to assess temporal consistency without the delays associated with extended time periods.

E. ALGORITHMIC IMPLEMENTATION FOR $R_{OVERLAP}$

The calculation of RO variations follows this systematic procedure:

Algorithm 1 Calculation of RO Variations

Require: Repetition factor rf, Query q
Ensure: RO_{minmin}, RO_{minmax}, RO_{ave}, and RO-Diff

```
Initialize RO ← ∅
for i = 1 to rf do
    Submit query q and store returned web pages in A_i
    for j = i + 1 to rf do
        Submit query q again and store returned web pages in A_j
        Compute R_overlap between A_i and A_j using Equation (20)
        RO ← RO ∪ {R_overlap}
    end for
end for
RO_minmin ← min(RO)
RO_minmax ← max(RO)
RO_ave ← (1/|RO|) Σ_{Y∈RO} Y
RO-Diff ← max_{Y∈RO} |Y − RO_minmin|
```

V. THE $R_{NON\text{-}OVERLAP}$ COMPONENT

In web search engine evaluation, comparing result sets A and B often reveals varying degrees of overlap. When these sets are not identical or only partially overlap, traditional metrics like R_{overlap} become inadequate, as they can only analyze common elements between sets. This limitation becomes particularly problematic when evaluating the ranking behavior of non-common web pages.

We introduce $R_{\text{non-overlap}}$, a comprehensive metric designed to analyze ranking patterns of unique web pages that appear exclusively in one result set. This metric requires at least one non-common web page between sets A and B to be applicable. When sets are completely identical, $R_{\text{non-overlap}}$ is undefined and defaults to zero.

A. FORMAL DESIGN OF $R_{NON\text{-}OVERLAP}$

For any query, a search engine retrieves m relevant web pages, though typically only the first n results are displayed due to pagination constraints. Our analysis focuses on these top n results, effectively transforming an unbounded result set into a finite one through truncation.

In the context of repeated queries, we consider a web page relevant if it appears in at least one of the retrieved result sets. The design of $R_{\text{non-overlap}}$ specifically targets the analysis of unique web pages that exist in one set but not the other.

1) RANK ASSIGNMENT AND MATHEMATICAL FORMULATION

Let us consider a web page X that exists in set A but not in set B. While X has a well-defined rank $r_A(X)$ in set A, its rank in set B is undefined. We postulate that X could have a rank in B anywhere between $n+1$ and m, represented as:

$$(n+1) \le r_B(X) \le m \tag{26}$$

This fundamental assumption leads to our formulation of $R_{\text{non-overlap}}$:

$$R_{\text{non-overlap}} = \frac{\sum\limits_{1 \le i \le |A-B|} \sum\limits_{X \in A-B} \frac{(n+i)-r_A(X)}{n}}{|A-B|} \times \frac{|A \cap B|}{n} \tag{27}$$

where:

- $A - B$ represents the set of web pages that appear in set A but not in set B
- $r_A(X)$ is the rank position of web page X in set A
- n is the maximum number of results considered
- $|A \cap B|$ is the number of common results between sets A and B

The first term quantifies the positional disparity of unique elements, while the second term weights this by the proportional overlap between sets.

B. BOUNDARY ANALYSIS OF $R_{NON\text{-}OVERLAP}$

We analyze the range of $R_{\text{non-overlap}}$ under two boundary conditions:

1) LOWER BOUND

When sets A and B share no common elements ($A \cap B = \emptyset$), then $|A \cap B| = 0$. Substituting into Equation (27) yields $R_{\text{non-overlap}} = 0.0$.

2) UPPER BOUND

When set A contains precisely one element not present in B ($|A - B| = 1$) and this element appears at the highest rank ($r_A(X) = 1$), we can simplify Equation (27):

$$\begin{aligned} R_{\text{non-overlap}} &= \frac{\frac{(n+1)-1}{n}}{1} \times \frac{|A \cap B|}{n} \\ &= \frac{n}{n} \times \frac{|A \cap B|}{n} \\ &= \frac{|A \cap B|}{n} \end{aligned} \tag{28}$$

As n increases and $|A \cap B|$ approaches n, $R_{\text{non-overlap}}$ approaches 1.0. Therefore:

$$0.0 \le R_{\text{non-overlap}} \le 1.0 \tag{29}$$

An optimal ranking algorithm minimizes $R_{\text{non-overlap}}$, while a suboptimal algorithm produces higher values. This behavior aligns with the intuition that effective search engines should maintain consistent rankings across repeated queries.

C. STATISTICAL VARIATION AND ALGORITHMIC COMPUTATION OF $R_{NON\text{-}OVERLAP}$

When queries are repeated multiple times with repetition factor rf, we observe $\binom{rf}{2}$ possible $R_{\text{non-overlap}}$ values. To systematically analyze this variation, we first determine

RN_{min}, the minimum observed $R_{\text{non-overlap}}$ value, and then compute the deviation set:

$$RN_{\text{Diff}} = \{|RN_{\text{min}} - Y|; \forall Y \in RN, Y \neq RN_{\text{min}}\} \tag{30}$$

From this set, we derive three key statistical measures:

$$\begin{aligned} RN_{\text{minmin}} &= \min(RN_{\text{Diff}}) \\ RN_{\text{minmax}} &= \max(RN_{\text{Diff}}) \\ RN_{\text{ave}} &= \frac{1}{|RN_{\text{Diff}}|} \sum_{Z \in RN_{\text{Diff}}} Z \end{aligned} \tag{31}$$

These metrics collectively quantify the stability and consistency of the ranking algorithm under repeated queries.

D. ALGORITHMIC IMPLEMENTATION FOR $R_{\text{NON-OVERLAP}}$

Algorithm 2 outlines the steps to determine RN_{Diff} and its derived statistics.

Algorithm 2 RN_{Diff} Calculation Procedure

Require: Repetition factor rf, Number of results n
Ensure: $RN_{\text{min}}, RN_{\text{Diff}}, RN_{\text{minmin}}, RN_{\text{minmax}}, RN_{\text{ave}}$

Initialize RN as a matrix of size $rf \times rf$
for $i = 1$ to rf **do**
  Execute query and retrieve the top n web pages as set A_i
  for $j = i + 1$ to rf **do**
    Execute query again and retrieve the top n web pages as set A_j
    $RN[i][j] = \frac{\sum_{1 \le l \le |A_i - A_j|} \sum_{X \in A_i - A_j} \frac{(n+l) - r_{A_i}(X)}{n}}{|A_i - A_j|} \times \frac{|A_i \cap A_j|}{n}$
  end for
end for
$RN_{\text{min}} = \min\{RN[i][j] > 0 : 1 \le i < j \le rf\}$
$RN_{\text{Diff}} = \{|RN_{\text{min}} - RN[i][j]| : RN[i][j] \neq RN_{\text{min}}, 1 \le i < j \le rf\}$
$RN_{\text{minmin}} = \min(RN_{\text{Diff}})$
$RN_{\text{minmax}} = \max(RN_{\text{Diff}})$
$RN_{\text{ave}} = \frac{1}{|RN_{\text{Diff}}|} \sum_{Z \in RN_{\text{Diff}}} Z$
return $RN_{\text{min}}, RN_{\text{Diff}}, RN_{\text{minmin}}, RN_{\text{minmax}}, RN_{\text{ave}}$

E. IMPLICATIONS FOR SEARCH ENGINE EVALUATION

The $R_{\text{non-overlap}}$ metric addresses a crucial evaluation gap by providing meaningful analysis of non-common search results. This approach offers several advantages for comprehensive search engine assessment:

- It captures information about ranking behavior that traditional overlap-only metrics miss
- It provides a mathematical framework for evaluating result consistency even when different pages are returned
- It enables fair comparison between search engines with different retrieval strategies
- It produces meaningful statistics that can identify unstable ranking algorithms

By combining R_{overlap} with $R_{\text{non-overlap}}$, evaluators gain a more complete picture of search engine performance across both common and unique results. This holistic approach more accurately reflects the user experience, where both the consistency of familiar results and the relevance of unique discoveries matter.

VI. $R_{\text{COMPREHENSIVE}}$: A UNIFIED APPROACH TO RANKING CONSISTENCY

When evaluating search engine performance across multiple queries, we need a holistic metric that considers both overlapping and non-overlapping results. While R_{overlap} quantifies consistency for common web pages and $R_{\text{non-overlap}}$ measures behavior for unique results, these metrics in isolation provide only partial insights into overall ranking performance. To address this limitation, we introduce $R_{\text{comprehensive}}$, a unified framework that integrates both perspectives.

A. FORMAL DEFINITION OF $R_{\text{COMPREHENSIVE}}$

The $R_{\text{comprehensive}}$ metric combines R_{overlap} and $R_{\text{non-overlap}}$ using a weighted approach:

$$R_{\text{comprehensive}} = \alpha \times R_{\text{overlap}} + (1 - \alpha) \times R_{\text{non-overlap}} \tag{32}$$

where:

- α is a weighting factor in the range [0, 1] that determines the relative importance of each component
- R_{overlap} quantifies ranking consistency for common results as defined in Equation (20)
- $R_{\text{non-overlap}}$ measures ranking behavior for unique results as defined in Equation (27)

This formulation ensures that $R_{\text{comprehensive}}$ maintains the same interpretable range as its component metrics:

$$0.0 \le R_{\text{comprehensive}} \le 1.0 \tag{33}$$

As with the individual metrics, lower values of $R_{\text{comprehensive}}$ indicate greater ranking consistency, with 0.0 representing perfect stability and 1.0 indicating maximum inconsistency.

B. PARAMETER SELECTION AND ADAPTATION

The weighting parameter α provides flexibility in emphasizing different aspects of ranking behavior based on evaluation priorities:

- $\alpha = 0.5$: Equal weighting between overlap and non-overlap components, suitable for general evaluation scenarios where both aspects are equally important
- $\alpha > 0.5$: Greater emphasis on ranking consistency of common results, appropriate for contexts where users are likely to focus on familiar web pages
- $\alpha < 0.5$: Greater emphasis on handling of unique results, beneficial when evaluating search engines' ability to introduce diverse or novel content

C. JUSTIFICATION OF WEIGHTING STRATEGY AND HANDLING OF UNCERTAINTY

The proposed *Rcomprehensive* metric combines *Roverlap* and *Rnon-overlap* using a convex linear combination controlled by the weighting factor $\alpha \in [0, 1]$:

$$R_{\text{comprehensive}} = \alpha \cdot R_{\text{overlap}} + (1 - \alpha) \cdot R_{\text{non-overlap}}$$

In our current implementation, the value of α is set to 0.5, implying an equal emphasis on overlapping and non-overlapping ranking behavior. This default choice is intended as a balanced baseline to initiate comparison. However, we acknowledge that this selection lacks a formal theoretical justification. To evaluate the effect of the weighting factor, we conducted a sensitivity analysis by varying α from 0.1 to 0.9 in increments of 0.1. The experiments were repeated across the complete set of Googlewhack and biographical queries for both Google and Bing.

Our observations indicate that in high-overlap scenarios, such as those associated with Googlewhack queries, the value of $R_{\text{comprehensive}}$ remains relatively stable across changes in α. However, in lower-overlap cases—especially for high-volume queries on Google—the metric demonstrates significant sensitivity to the value of α. This finding suggests that an adaptive or data-driven approach to determining the weighting factor may be more appropriate than a static assignment. As a future extension, we propose learning α via query-specific optimization techniques, such as cross-validation on user feedback, minimization of empirical rank deviation, or entropy-based heuristics that reflect the relative information certainty in overlapping versus non-overlapping result segments.

Additionally, the current formulation treats all ranking observations as deterministic and does not explicitly incorporate uncertainty, despite the stochastic nature of web search outputs caused by personalization, load balancing, and algorithm experimentation. To enhance robustness, the incorporation of uncertainty-aware fusion strategies is a promising direction. This may involve the use of fuzzy integrals, such as the Choquet or Sugeno integrals, which allow interaction modeling under vagueness, or evidence-based methods such as Dempster-Shafer theory to express confidence levels in rank agreement. Interval-based or probabilistic ranking formulations could also be employed, allowing each ranked position to be represented as a range or distribution rather than a fixed value. These extensions would enable Rmeasure to model not only structural rank variation but also the degree of certainty associated with each ranking instance, thereby strengthening its applicability in real-world, noisy search environments.

D. STATISTICAL EXTENSIONS FOR REPEATED QUERIES

When evaluating search engine performance across repeated queries with repetition factor rf, we extend $R_{\text{comprehensive}}$ to capture statistical variations. The combined statistical measures are derived from the corresponding metrics of the component parts:

$$RC_{\text{minmin}} = \alpha \times RO_{\text{minmin}} + (1 - \alpha) \times RN_{\text{minmin}} \tag{34}$$

$$RC_{\text{minmax}} = \alpha \times RO_{\text{minmax}} + (1 - \alpha) \times RN_{\text{minmax}} \tag{35}$$

$$RC_{\text{ave}} = \alpha \times RO_{\text{ave}} + (1 - \alpha) \times RN_{\text{ave}} \tag{36}$$

where:

- RO_{minmin}, RO_{minmax}, RO_{ave} represent the statistical metrics derived from R_{overlap} values
- RN_{minmin}, RN_{minmax}, RN_{ave} represent the corresponding metrics from $R_{\text{non-overlap}}$ values

These combined measures provide a comprehensive view of ranking consistency across both overlapping and non-overlapping results. To further quantify the stability of $R_{\text{comprehensive}}$ across repeated queries, we define the deviation measure:

$$RC\text{-Diff} = \{|RC_{\text{min}} - Y| \mid Y \in RC, Y \neq RC_{\text{min}}\} \tag{37}$$

where RC represents the set of all $R_{\text{comprehensive}}$ values calculated across multiple query repetitions, and RC_{min} is the minimum value in this set.

E. ALGORITHMIC IMPLEMENTATION FOR $R_{COMPREHENSIVE}$

Algorithm 3 outlines the systematic procedure for calculating $R_{\text{comprehensive}}$ and its statistical variations:

Algorithm 3 Calculation of $R_{\text{comprehensive}}$ Variations

Require: Repetition factor rf, Query q, Weight α
Ensure: RC_{minmin}, RC_{minmax}, RC_{ave}, and RC-Diff

Initialize $RC \leftarrow \emptyset$
for $i = 1$ to rf **do**
  Submit query q and store returned web pages in A_i
  for $j = i + 1$ to rf **do**
    Submit query q again and store returned web pages in A_j
    Compute R_{overlap} between A_i and A_j using Equation (20)
    Compute $R_{\text{non-overlap}}$ between A_i and A_j using Equation (27)
    $R_{\text{comprehensive}} \leftarrow \alpha \times R_{\text{overlap}} + (1 - \alpha) \times R_{\text{non-overlap}}$
    $RC \leftarrow RC \cup \{R_{\text{comprehensive}}\}$
  end for
end for
$RC_{\text{min}} \leftarrow \min(RC)$
$RC_{\text{minmin}} \leftarrow \min\{|RC_{\text{min}} - Y| \mid Y \in RC, Y \neq RC_{\text{min}}\}$
$RC_{\text{minmax}} \leftarrow \max\{|RC_{\text{min}} - Y| \mid Y \in RC, Y \neq RC_{\text{min}}\}$
$RC_{\text{ave}} \leftarrow \frac{1}{|RC|} \sum_{Y \in RC} Y$
RC-Diff $\leftarrow \{|RC_{\text{min}} - Y| \mid Y \in RC, Y \neq RC_{\text{min}}\}$
return RC_{minmin}, RC_{minmax}, RC_{ave}, and RC-Diff

F. PRACTICAL APPLICATIONS AND SIGNIFICANCE

The $R_{\text{comprehensive}}$ framework offers several practical advantages for search engine evaluation:

- **Holistic Assessment**: By considering both overlapping and non-overlapping results, it provides a more complete picture of search engine performance than traditional metrics.
- **Flexible Evaluation**: The weighting parameter allows evaluators to customize the metric according to specific evaluation objectives or search contexts.
- **User-Centered Approach**: The adaptive weighting mechanism naturally aligns with user experiences, where the relative importance of common versus unique results varies based on the query context.
- **Statistical Robustness**: The extended statistical measures enable evaluators to quantify not only average performance but also best-case and worst-case scenarios, providing insights into ranking stability.

When applied to comparative search engine evaluation, $R_{\text{comprehensive}}$ can reveal subtle differences in ranking behavior that might be overlooked by more traditional metrics. For instance, two search engines might exhibit similar performance on overlapping results but differ significantly in how they handle unique web pages. The comprehensive framework captures these nuances, enabling more informed decisions about search engine quality and appropriate use cases.

Furthermore, the metric can guide algorithm development by highlighting specific areas for improvement—whether in maintaining consistency for common results or in optimizing the ranking of unique content. This targeted feedback mechanism can accelerate the refinement of ranking algorithms and ultimately enhance the user search experience.

VII. RESULTS AND DISCUSSIONS

The detail of the experimental setup, the experiments, and the results of the experiments are discussed in this section.

A. EXPERIMENTAL SETUP

The experiments entirely depend on the web-search engines, query, query repetition factor (rf), and query repetition method.

1) WEB-SEARCH ENGINES

The alphametic.com application was used to analyze the market share of web search engines in the top 15 GDP countries. According to this analysis, Google is in the top spot in 14 countries. Bing is in the next spot after Google, in 13 countries. Due to their prevalence of use in these countries, the experiments were conducted on the search engines Google and Bing.

2) QUERY REPETITION METHOD

In our experiments, the queries are repeated in two possible ways. The methods are: 1) Repeating the query as a whole (phrase-based repetition method), and 2) Split the query into tokens, and repeat the query as tokens (token-based repetition method). If a given query is a single word query, then these two query repetition methods become the same. Hence, a query should contain at least two words.

3) QUERIES

As the queries are repeated, and the performance analysis is depending on the number of relevant web-pages returned by the repeated queries, the type and nature of the queries play an important role. Hence, two sets of queries are used in the experiments.

The queries which will fetch a minimum number of hits from the web-search engines form the first set. A Google search query that only returns a single entry in the search has become known as a Googlewhack [24]. As the experiment needs a query with a low number of results, the use of a Googlewhack is a useful option. Five queries from the Googlewhack list are selected. The selected queries are: 1) ambidextrous scallywags, 2) anxiousness scheduler, 3) assonant octosyllable, 4) bamboozle guzzler, and 5) illuminatus ombudsman [25].

The second set consists of high-volume search queries that consistently generate substantial search results. These well-established biographical queries represent notable public figures with significant digital footprints, providing an effective contrast to the minimal-result Googlewhack queries. The selected queries are: 1) anthony bourdain, 2) kate spade, 3) meghan markle, 4) stan lee, and 5) stephen hawking [26].

4) QUERY REPETITION FACTOR (RF)

Trial experiments were conducted, and the overlap between the successive repetition factors was analyzed. Based on the results, it is confirmed that the overlap is maximum when a query is repeated for '4', and '5' times. Hence '5' is selected as the maximum value for the repetition factor.

5) NUMBER OF RELEVANT WEB-PAGES (N)

We restricted the number of relevant web-pages (n) to 120, as the users are interested only in the first few result pages. The number of relevant web-pages (n) is varied in a step of 10. Hence, there are 12 possible levels of relevant web-pages (10, 20, …, 120), and the experiments are conducted at all these levels.

B. RESULTS, AND ANALYSIS OF ROVERLAP

Table 2 shows the Roverlap values of Bing and Google for the two sets of queries. In the table, n represents the document levels which we increased in steps of 10. The labels 'Tokens' and 'Phrase' represent the token-based and phrase-based repetition methods respectively.

From Table 2, it is identified that the phrase and token-based repetition methods in Googlewhack and high-volume biographical queries produce almost the same level of performance in Bing. Whereas in Google, there is a significant difference between the phrase and token-based repetition methods in high-volume biographical queries. There are no such differences in Googlewhack queries.

From the above discussion, it is identified that irrespective of the type of repetition methods, Bing gives almost the same rank or position to the common or overlapping web-pages. Google, however, gives a different rank or position to the overlapping web-pages when the number of hits to a query is maximum. Interestingly, Google gives almost the same position for phrase and token-based repetition method when the number of hits of a query is minimum. An average search engine user can use either phrase or token-based repetition methods to receive a better performance in Bing. In Google, however, the user has to use both token and phrase-based methods for getting better performance if the number of hits of a query is not known.

C. AVERAGE RANK DIFFERENCE ANALYSIS

The $R_{overlap}$ metric introduced in Section IV-B serves as an indirect indicator of rank differences among relevant or overlapping web pages returned by search engines for repeated queries. Since $R_{overlap}$ depends on $A \cap B$, $A \cup B$, and rank differences—with rank difference being the most critical factor and directly proportional to $R_{overlap}$—we conducted a separate analysis of rank difference values.

The formula used for calculating $R_{overlap}$ in Equation (20) can be rearranged to calculate the average rank difference value:

$$\text{Avg. Rank Diff.} = \left[\frac{2nx(n-1) - x^2(n-1)}{n}\right] \cdot R_{overlap} \tag{38}$$

where $x = |A \cap B|$ represents the cardinality of the intersection between result sets. This formulation reveals an inverse relationship between $|A \cap B|$ and the average rank difference—as the intersection of results increases, the average rank difference decreases. This demonstrates a fundamental trade-off between result overlap and ranking consistency.

Table 3 presents the average rank difference values for both Googlewhack queries and High-volume biographical queries across Bing and Google search engines.

Our analysis reveals that the patterns in average rank difference values closely align with those observed in the $R_{overlap}$ analysis. For Google, High-volume Biographical queries exhibit significantly higher average rank differences compared to Googlewhack queries. Notably, for queries with minimal hits, the variation in average rank difference becomes negligible across different query repetition methods. In contrast, Bing demonstrates more consistent rankings for relevant web pages regardless of the query repetition approach employed.

This analysis provides quantitative evidence of the ranking consistency characteristics of these major search engines. Google shows greater sensitivity to query volume, while Bing exhibits more stable ranking behavior across repeated queries. These findings have important implications for search engine evaluation and optimization, particularly in understanding how ranking algorithms respond to different types of queries and repetition methods.

D. RESULTS AND ANALYSIS OF RO_{MINMIN}, RO_{MINMAX}, AND RO_{AVE}

The $R_{overlap}$ values are calculated at various repetition factor levels. The variation of $R_{overlap}$ values against repetition factor (rf) variations cannot be easily correlated. When correlated, this produces a set of values which we have condensed into single metrics for clearer analysis. We use RO_{minmin}, RO_{minmax}, and RO_{ave} for this purpose. The RO_{minmin} gives the minimum range of $R_{overlap}$ value when the rf value is varied. Similarly, RO_{minmax} gives the maximum range, and RO_{ave} gives the average range of $R_{overlap}$ variation when the rf value is varied.

Tables 4 and 5 present the RO_{minmin}, RO_{minmax}, and RO_{ave} values for Bing and Google, respectively, for both Googlewhack and High-volume biographical queries.

The formula used to calculate $R_{overlap}$ was previously discussed in Equation (20). The numerator of this formula contains two terms: i) n, and ii) rank difference of the common or overlapping ($A \cap B$) web pages. The denominator also contains two terms: i) $A \cap B$, and ii) $A \cup B$.

From the results presented in Tables 4 and 5, we can observe several patterns in the search engine behavior. For both Bing and Google, the maximum values of RO_{minmax} occur at lower values of N (typically $N = 10$ or $N = 20$), indicating greater variability in overlap metrics at smaller result set sizes. As N increases, both RO_{minmax} and RO_{ave} generally decrease, suggesting more stable overlap patterns with larger result sets.

When comparing search engines, Google shows higher RO_{ave} values than Bing for Googlewhack queries, indicating greater variation in result rankings when queries have minimal hits. For High-volume biographical queries, the pattern becomes more complex, with Google maintaining more consistent RO_{minmin} values across different values of N compared to Bing.

These metrics provide valuable insight into the stability and consistency of search engine results when queries are repeated under varying conditions, revealing fundamental differences in the ranking algorithms employed by Bing and Google.

Assume that at a particular level of 'n' and 'rf', the value of $A \cap B$ for a query is x times higher when compared to another query repetition factor. As a result, the numerator value gets increased by a factor of y, provided that the common web-pages don't occupy the same position. As $|A \cap B|$ increases, $|A \cup B|$ automatically decreases. Consequently, the product of $|A \cup B|$ and $|A \cap B|$ is also increased by a factor of z. However, the value of y is greater than the value of z.

Hence, at a particular level of 'n' and 'rf', if the value of $|A \cap B|$ is high for a particular query, it leads to the $R_{overlap}$ value's variation. As $R_{overlap}$ value varies, the minimum, maximum, and average range (RO_{minmin}, RO_{minmax}, and RO_{ave}) also vary. We represented this variation graphically

TABLE 2. Roverlap Values of bing and Google Search engines using different query types and repetition methods.

(a) Googlewhack Queries					(b) High-volume biographical queries				
n	Bing		Google		n	Bing		Google	
	Phrase	Tokens	Phrase	Tokens		Phrase	Tokens	Phrase	Tokens
10	0.0823	0.103	0.1177	0.1209	10	0.1586	0.1329	0.1215	0.1511
20	0.1115	0.1103	0.0986	0.1021	20	0.14	0.1559	0.1248	0.1162
30	0.108	0.1072	0.0844	0.0859	30	0.141	0.1533	0.1211	0.1225
40	0.0994	0.1022	0.0717	0.0877	40	0.1403	0.1477	0.1028	0.1211
50	0.0966	0.1017	0.0641	0.0839	50	0.141	0.1436	0.0942	0.1193
60	0.0964	0.1046	0.06	0.0796	60	0.1345	0.1408	0.0888	0.1198
70	0.0904	0.0975	0.0577	0.0779	70	0.137	0.1389	0.0824	0.1251
80	0.0966	0.1016	0.0551	0.0769	80	0.1434	0.139	0.0742	0.1161
90	0.0895	0.0999	0.0552	0.0747	90	0.142	0.1396	0.0708	0.1195
100	0.088	0.0943	0.0535	0.0726	100	0.1393	0.1358	0.068	0.1185
110	0.0939	0.0924	0.0533	0.071	110	0.1354	0.1341	0.0656	0.114
120	0.0904	0.0925	0.0527	0.0705	120	0.1334	0.1334	0.0634	0.113

TABLE 3. Average rank difference values of Bing and Google.

(a) Googlewhack Queries					(b) High-volume biographical queries				
n	Bing		Google		n	Bing		Google	
	Phrase	Tokens	Phrase	Tokens		Phrase	Tokens	Phrase	Tokens
10	0.926	1.285	1.452	1.52	10	2.041	1.811	1.489	2.112
20	2.559	2.578	2.43	2.664	20	3.549	4.129	3.101	3.511
30	3.902	3.986	2.966	3.238	30	5.347	6.13	4.416	5.579
40	4.853	5.096	3.275	4.418	40	7.179	7.818	5.091	7.351
50	5.945	6.335	3.598	5.162	50	9.086	9.522	6.018	9.149
60	7.178	7.99	3.927	5.741	60	10.463	11.172	6.871	11.141
70	7.942	8.731	4.378	6.395	70	12.414	13.001	7.417	13.625
80	9.779	10.474	4.699	7.114	80	14.744	14.923	7.713	14.413
90	10.321	11.542	5.22	7.628	90	16.348	16.89	8.328	16.509
100	11.253	12.116	5.456	8.047	100	17.974	18.343	8.899	17.985
110	13.607	13.035	5.888	8.448	110	19.327	20.012	9.364	18.856
120	14.285	14.312	6.242	9.037	120	20.797	21.622	9.815	20.211

TABLE 4. RO_{minmin}, RO_{minmax}, and RO_{ave} Values of Bing.

N	Googlewhack Queries						High-volume Biographical Queries					
	Phrase			Tokens			Phrase			Tokens		
	RO_{minmin}	RO_{minmax}	RO_{ave}	RO_{minmin}	RO_{minmax}	RO_{ave}	RO_{minmin}	RO_{minmax}	RO_{ave}	RO_{minmin}	RO_{minmax}	RO_{ave}
10	0.0196	0.109	0.0571	0.0513	0.1691	0.1011	0.0357	0.2551	0.1372	0.033	0.211	0.0992
20	0.0189	0.1549	0.0752	0.0121	0.1398	0.0655	0.0411	0.1701	0.0924	0.0173	0.1528	0.0739
30	0.0256	0.1377	0.0766	0.0206	0.1241	0.0678	0.0223	0.1174	0.0665	0.0159	0.1243	0.0684
40	0.0255	0.1343	0.077	0.0153	0.1272	0.0673	0.004	0.106	0.0472	0.0245	0.1277	0.0685
50	0.0199	0.1197	0.0641	0.0117	0.1132	0.0571	0.0186	0.1157	0.0552	0.0244	0.0988	0.059
60	0.0179	0.109	0.0607	0.0114	0.1138	0.0565	0.016	0.0857	0.0431	0.0309	0.0963	0.0607
70	0.0182	0.0962	0.0561	0.0099	0.1065	0.0501	0.0163	0.0956	0.0515	0.0223	0.1004	0.0574
80	0.0157	0.0974	0.0594	0.0108	0.1066	0.0559	0.0175	0.1092	0.057	0.022	0.0958	0.0598
90	0.0127	0.0909	0.0531	0.0101	0.1014	0.0554	0.0142	0.1078	0.0553	0.0245	0.1019	0.0615
100	0.0085	0.0843	0.0481	0.0181	0.0958	0.0562	0.0177	0.099	0.0501	0.0202	0.095	0.0551
110	0.0163	0.0959	0.0511	0.013	0.1008	0.0575	0.0152	0.0942	0.0493	0.0153	0.0954	0.0511
120	0.0137	0.0925	0.0472	0.0126	0.1056	0.0506	0.0102	0.0928	0.0448	0.0203	0.1011	0.0502

as a minmax graph. Figures 1 and 2 show the variation of RO_{minmin}, RO_{minmax}, and RO_{ave} of Bing and Google for both types of repetition methods.

The range variation illustrates the variation in the $R_{overlap}$ values. The exact range variation and its values are analyzed using the area under the curve analysis. The phrase-based repetition method in Bing has 9.309, and the token-based method has 10.515 as the area under the curve values. In Google, the phrase-based method has 9.629, and the token-based method has 10.45 as the area under the curve value. From the area under the curve analysis, it is confirmed that the variation of the overlapping web-page's rank is almost similar in both token and phrase-based repetition methods for both Bing and Google.

E. RESULT AND ANALYSIS OF $R_{NON-OVERLAP}$

The number of overlapping web-pages and their rank variations are analyzed in $S_{measure}$ and $R_{overlap}$. When a query is repeated and submitted to a web-search engine, the engine typically returns both overlapping and non-overlapping

TABLE 5. RO_{minmin}, RO_{minmax}, and RO_{ave} values of Google.

N	Googlewhack Queries						High-volume Biographical Queries					
	Phrase			Tokens			Phrase			Tokens		
	RO_{minmin}	RO_{minmax}	RO_{ave}	RO_{minmin}	RO_{minmax}	RO_{ave}	RO_{minmin}	RO_{minmax}	RO_{ave}	RO_{minmin}	RO_{minmax}	RO_{ave}
10	0.039	0.185	0.1078	0.0464	0.2052	0.1303	0.0327	0.1539	0.0987	0.0581	0.2655	0.1493
20	0.0454	0.1598	0.0995	0.0358	0.1455	0.0857	0.0256	0.1324	0.0751	0.0163	0.1263	0.066
30	0.0348	0.1393	0.0864	0.0261	0.1251	0.0722	0.0137	0.1168	0.0694	0.0188	0.1023	0.054
40	0.028	0.1248	0.0726	0.0246	0.129	0.0761	0.0142	0.1103	0.0643	0.0225	0.0965	0.0567
50	0.0215	0.1136	0.0644	0.022	0.1172	0.0698	0.0174	0.1016	0.0602	0.0215	0.1119	0.067
60	0.02	0.1088	0.0609	0.0173	0.1075	0.0645	0.0175	0.0968	0.0564	0.0167	0.1084	0.0632
70	0.018	0.1098	0.0589	0.0172	0.1056	0.0642	0.0171	0.087	0.0514	0.0178	0.1159	0.0713
80	0.0173	0.1053	0.0564	0.0167	0.1062	0.0611	0.0179	0.0744	0.0452	0.0245	0.106	0.065
90	0.0166	0.1072	0.0567	0.0142	0.1043	0.0582	0.0141	0.0743	0.0433	0.0211	0.1077	0.0691
100	0.0148	0.1057	0.0546	0.014	0.101	0.0563	0.0136	0.0693	0.0395	0.0203	0.1116	0.0714
110	0.0143	0.1068	0.0546	0.0156	0.1002	0.0571	0.0125	0.0686	0.0388	0.0237	0.0998	0.0679
120	0.014	0.1056	0.054	0.0158	0.1011	0.0576	0.0126	0.068	0.0392	0.0231	0.109	0.0678

web-pages. While $S_{measure}$ directly analyzes the number of overlapping web-pages and indirectly addresses non-overlapping web-pages, the rank of non-overlapping web-pages requires further examination. The $R_{non-overlap}$ metric addresses this gap by analyzing the rank of non-common or non-overlapping web-pages.

The formula used to calculate $R_{non-overlap}$ was previously discussed in equation (27). Two sets of queries were repeatedly submitted to Bing and Google at various 'rf' levels, and the $R_{non-overlap}$ value was calculated for these repeated queries. Table 6 presents the $R_{non-overlap}$ values of Bing and Google at various 'n' levels.

Analysis of Table 6 reveals that the query repetition method significantly impacts $R_{non-overlap}$ values in both Bing and Google. However, for High-volume Biographical Queries in Bing, the difference in $R_{non-overlap}$ values between phrase-based and token-based repetition methods is minimal. These findings confirm that repeating queries using both phrase-based and token-based repetition methods can enhance the performance of web-search engines' ranking mechanisms.

The $R_{non-overlap}$ values for both Bing and Google show a decreasing trend as the 'n' value increases in both Googlewhack and High-volume Biographical Queries. Interestingly, when 'n' reaches 50 or 60, the $R_{non-overlap}$ value begins to increase. This minimum value indicates a significant presence of non-overlapping web-pages at these particular 'n' values. Therefore, when 'n' equals 50 or 60, repeating queries yields the maximum number of previously undiscovered web-pages.

1) IMPACT OF THE NON-OVERLAPPING WEB-PAGE'S RANK

The rank of non-overlapping web-pages directly influences $R_{non-overlap}$ values. When non-overlapping web-pages have lower ranks (appearing higher in search results), the $R_{non-overlap}$ value increases. Conversely, when these pages have higher ranks (appearing lower in search results), the $R_{non-overlap}$ value decreases. To verify and analyze this relationship between non-common web-page ranks and $R_{non-overlap}$ values, we conducted further investigation.

The total number of web-pages 'n' was divided into three segments: i) first thirty percent (0-30%), ii) middle forty percent (31)-70%), and iii) last thirty percent (71-100%). For example, with an 'n' value of 10, these segments correspond to positions 1-3, 4-7, and 8-10, respectively. After dividing the web-pages into these segments, we calculated the percentage of non-overlapping web-pages within each segment. This calculation was performed for both phrase-based and token-based repetition methods in Bing and Google. Figures 3 and 4 illustrate the results of this analysis.

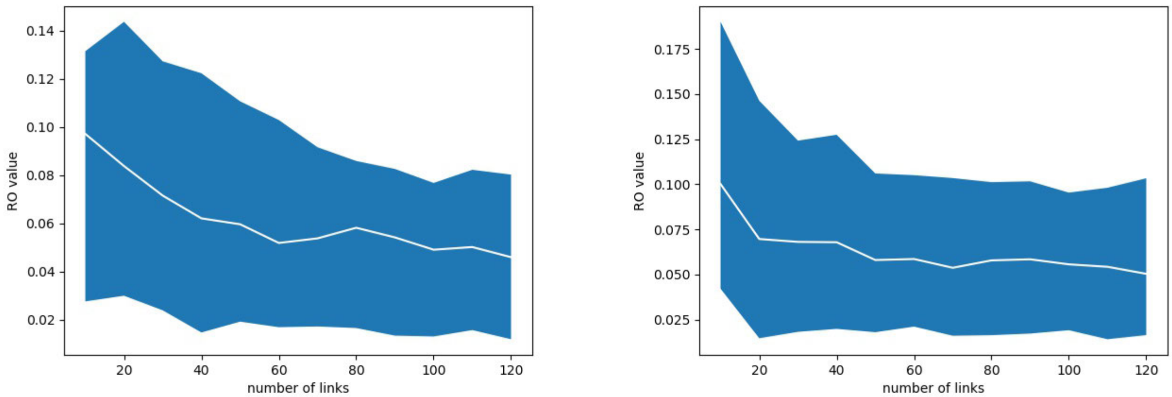
Figures 3 and 4 demonstrate that as the 'n' value increases, the percentage of non-overlapping web-pages in the first thirty percent gradually increases for both phrase-based and token-based repetition methods in Bing and Google. The proportion of non-overlapping web-pages in the middle forty percent is lower compared to the last thirty percent at small values of 'n', but this relationship reverses at higher 'n' values.

Having calculated the percentage distribution of non-overlapping web-pages across these three segments at various 'n' values, we analyzed the correlation between these percentages and the corresponding $R_{non-overlap}$ values.

Our analysis revealed a negative correlation between the percentage of non-overlapping web-pages in the first thirty percent and $R_{non-overlap}$ values, confirming our earlier hypothesis. Non-common web-pages appearing at the top of search results tend to reduce $R_{non-overlap}$ values. For a high-performing search engine, the $R_{non-overlap}$ value should be minimal. Therefore, an effective search engine should have a high percentage of non-overlapping web-pages in the first thirty percent of results. In other words, non-overlapping web-pages should ideally occupy prominent positions in web-search engine results.

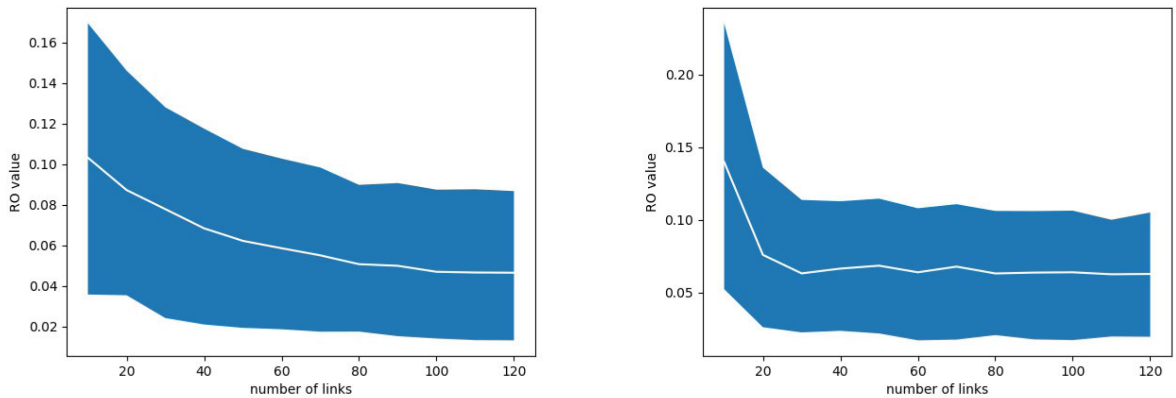
F. RESULTS AND ANALYSIS OF RN_{MINMIN}, RN_{MINMAX}, AND RN_{AVE}

The $R_{non-overlap}$ value's variation is directly related to the rank or position of the non-overlapping webpages. As the rank of the non-overlapping web-pages varies from a query to another, and from the phrase-based to the token-based repetition methods, we analyzed the range of the non-overlapping web-pages' rank variation. It can be indirectly analyzed by using the $R_{non-overlap}$ value. The minimum of



(a) phrase (b) tokens

FIGURE 1. ***$R_{overlap}$*** **range of Bing.**



(a) phrase (b) tokens

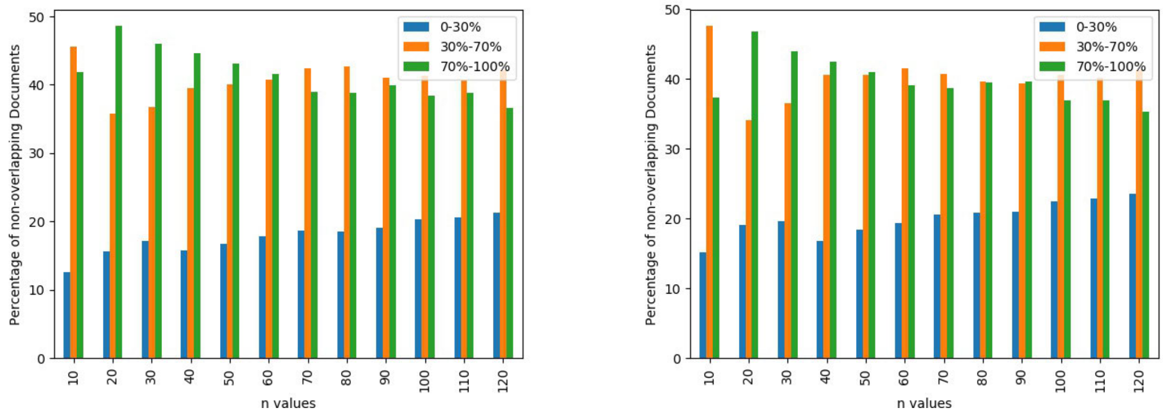
FIGURE 2. ***$R_{overlap}$*** **range of Google.**

TABLE 6. ***$R_{non-overlap}$*** **values of Bing and Google.**

n	Googlewhack queries				High-volume Biographical Queries			
	Bing		Google		Bing		Google	
	Phrase	Tokens	Phrase	Tokens	Phrase	Tokens	Phrase	Tokens
10	0.2474	0.257	0.3125	0.2775	0.3295	0.2899	0.3407	0.2987
20	0.2299	0.2461	0.3594	0.2856	0.2678	0.2582	0.3491	0.2691
30	0.2279	0.2372	0.3397	0.281	0.2557	0.2539	0.3592	0.2824
40	0.2313	0.2327	0.3317	0.2744	0.2491	0.2455	0.3374	0.2846
50	0.2322	0.2363	0.3276	0.2706	0.2445	0.2446	0.3267	0.2892
60	0.2401	0.2488	0.3082	0.2721	0.2406	0.2381	0.3411	0.2928
70	0.2465	0.2521	0.3014	0.2634	0.239	0.2319	0.3617	0.2945
80	0.2426	0.2483	0.2955	0.2586	0.2427	0.2308	0.3568	0.2998
90	0.2384	0.249	0.2901	0.2514	0.2433	0.2335	0.3653	0.3102
100	0.2349	0.244	0.2796	0.2544	0.2385	0.2309	0.3609	0.3121
110	0.2324	0.2411	0.2654	0.2517	0.2367	0.23	0.3578	0.312
120	0.2321	0.2364	0.2515	0.2379	0.2362	0.231	0.354	0.3106

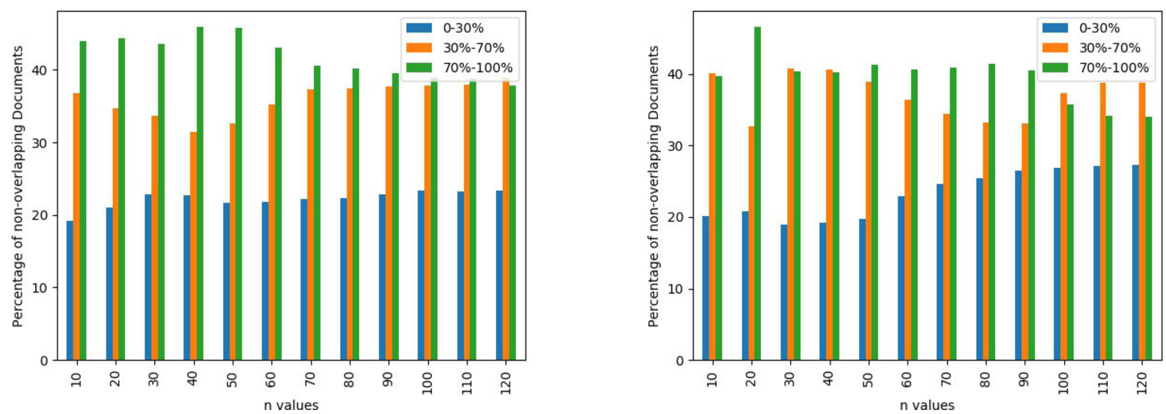
the $R_{non-overlap}$ at various 'n' level is calculated, and based on the minimum value, the range of the $R_{non-overlap}$ value is calculated. The minimum deviation is called RN_{minmin}, the maximum deviation is called RN_{minmax}, and the average deviation is termed RN_{ave}. Tables 7 and 8 show the RN_{minmin}, RN_{minmax}, and RN_{ave} values for Googlewhack and High-volume Biographical Queries for both Bing and Google.

The RN_{minmin}, RN_{minmax}, and RN_{ave} variations are almost the same for both phrase and token-based repetition methods in Bing and Google. Using the Googlewhack queries category in Google, the value of RN_{minmin} in token-based and



(a) phrase (b) tokens

FIGURE 3. Percentage of non-overlapping web pages at various 'n' levels in Bing.



(a) phrase (b) tokens

FIGURE 4. Percentage of non-overlapping web pages at various 'n' levels in Google.

TABLE 7. RN_{minmin}, RN_{minmax}, and RN_{ave} values of Bing.

N	Googlewhack Queries						High-volume Biographical Queries					
	Phrase			Tokens			Phrase			Tokens		
	RN_{minmin}	RN_{minmax}	RN_{ave}	RN_{minmin}	RN_{minmax}	RN_{ave}	RN_{minmin}	RN_{minmax}	RN_{ave}	RN_{minmin}	RN_{minmax}	RN_{ave}
10	0.094	0.298	0.181	0.086	0.297	0.187	0.028	0.228	0.118	0.044	0.246	0.145
20	0.020	0.224	0.103	0.043	0.284	0.144	0.015	0.182	0.087	0.032	0.173	0.090
30	0.031	0.259	0.122	0.050	0.237	0.134	0.015	0.156	0.078	0.028	0.139	0.083
40	0.042	0.228	0.126	0.045	0.201	0.120	0.014	0.120	0.061	0.019	0.132	0.073
50	0.031	0.191	0.101	0.038	0.178	0.107	0.008	0.110	0.059	0.017	0.145	0.078
60	0.027	0.177	0.097	0.025	0.145	0.091	0.023	0.126	0.071	0.010	0.132	0.075
70	0.028	0.188	0.094	0.020	0.170	0.098	0.016	0.109	0.060	0.008	0.110	0.063
80	0.020	0.142	0.074	0.017	0.152	0.088	0.014	0.097	0.055	0.015	0.102	0.065
90	0.017	0.139	0.069	0.021	0.174	0.095	0.014	0.106	0.061	0.009	0.106	0.062
100	0.006	0.139	0.066	0.017	0.166	0.086	0.010	0.102	0.055	0.007	0.097	0.059
110	0.006	0.138	0.065	0.017	0.166	0.086	0.010	0.096	0.058	0.013	0.107	0.062
120	0.008	0.156	0.069	0.017	0.156	0.086	0.006	0.098	0.057	0.019	0.109	0.068

phrase-based repetition methods has a significant variation. From the above analysis, it is confirmed that Google places the non-overlapping web-pages in a slightly different manner at the bottom of the results when the number of hits for those queries is minimum. Apart from that, both Bing and Google place the non-overlapping web-pages in the same manner.

TABLE 8. RN_{minmin}, RN_{minmax}, and RN_{ave} values of Google.

N	Googlewhack Queries						High-volume Biographical Queries					
	Phrase			Tokens			Phrase			Tokens		
	RN_{minmin}	RN_{minmax}	RN_{ave}	RN_{minmin}	RN_{minmax}	RN_{ave}	RN_{minmin}	RN_{minmax}	RN_{ave}	RN_{minmin}	RN_{minmax}	RN_{ave}
10	0.069	0.363	0.168	0.189	0.448	0.317	0.115	0.307	0.196	0.063	0.241	0.146
20	0.149	0.629	0.285	0.114	0.442	0.263	0.083	0.270	0.163	0.050	0.232	0.124
30	0.101	0.554	0.238	0.126	0.370	0.221	0.115	0.303	0.205	0.011	0.247	0.120
40	0.041	0.411	0.164	0.082	0.306	0.172	0.039	0.254	0.128	0.017	0.237	0.116
50	0.033	0.374	0.160	0.094	0.285	0.164	0.030	0.181	0.094	0.016	0.232	0.122
60	0.028	0.322	0.135	0.064	0.260	0.135	0.030	0.153	0.078	0.014	0.219	0.126
70	0.024	0.325	0.130	0.075	0.277	0.160	0.017	0.163	0.083	0.023	0.215	0.125
80	0.023	0.344	0.129	0.071	0.254	0.146	0.029	0.148	0.082	0.024	0.199	0.119
90	0.018	0.325	0.111	0.070	0.246	0.128	0.026	0.133	0.076	0.025	0.177	0.111
100	0.017	0.318	0.098	0.058	0.285	0.129	0.016	0.142	0.072	0.024	0.205	0.115
110	0.016	0.269	0.082	0.059	0.218	0.123	0.015	0.135	0.062	0.023	0.222	0.116
120	0.012	0.247	0.069	0.062	0.212	0.119	0.014	0.120	0.063	0.029	0.219	0.118

G. RESULTS AND ANALYSIS OF $R_{COMPREHENSIVE}$

The stability of search engine result rankings can be comprehensively evaluated by combining the non-overlap measurements with their range variations. We propose a new metric, $R_{comprehensive}$, which is calculated by taking 50% of the $R_{non-overlap}$ values and 50% of the range values previously analyzed. This comprehensive metric provides a holistic view of ranking stability, where values closer to 0 indicate highly stable rankings and values approaching 1 represent non-stable or highly variable rankings.

Table 9 presents the $R_{comprehensive}$ values for both Bing and Google search engines across Googlewhack queries and High-volume Biographical Queries. These values offer insights into the overall stability patterns of search engine rankings under different query complexities and search volumes.

Analysis of the $R_{comprehensive}$ values reveals several significant patterns in search engine ranking stability. For Googlewhack queries, Bing demonstrates consistently higher stability (lower $R_{comprehensive}$ values) compared to Google across most N values, particularly for phrase-based queries. This suggests that Bing maintains more consistent rankings for queries with minimal result sets. The stability difference between Bing and Google is most pronounced at lower N values (N=10 to N=30) and gradually converges as N increases, indicating that both search engines tend to stabilize their rankings as more results are considered.

For High-volume Biographical Queries, a different pattern emerges. Google's token-based approach demonstrates comparable stability to Bing's methods at lower N values, but as N increases beyond 70, Google's $R_{comprehensive}$ values consistently rise, indicating decreased stability for these biographical queries. This phenomenon may be attributed to Google's algorithm placing greater emphasis on semantic relevance for biographical information, leading to more variability in ranking patterns as the result set expands.

An interesting observation is that for both search engines, the phrase-based method generally shows higher stability (lower $R_{comprehensive}$ values) than the token-based method for Googlewhack queries, while the opposite trend is observed for High-volume Biographical Queries. This suggests that the optimal query processing strategy differs based on the query type and expected result volume.

The gradual convergence of $R_{comprehensive}$ values for Googlewhack queries as N increases (particularly in Google's results) indicates that ranking stability tends to improve with larger result sets for low-volume queries. Conversely, for High-volume Biographical Queries, we observe more consistent $R_{comprehensive}$ values across different N values, suggesting that ranking stability is less dependent on result set size for these high-volume queries.

These findings offer valuable insights for search engine optimization strategies, particularly for content targeting specific query types. Content optimized for high-volume biographical queries may benefit from token-based approaches on Bing, while phrase-based optimization might be more effective for low-volume, specialized queries across both search engines.

H. STATISTICAL ANALYSIS OF RANKING STABILITY METRICS

To assess the statistical significance of the differences observed between Bing and Google, we conducted paired t-tests on key ranking stability metrics across the query types and repetition methods.

For *Roverlap*, representing the consistency of overlapping results, paired comparisons on Googlewhack phrase queries ($N = 12$) showed a significant difference between Bing (mean = 0.096) and Google (mean = 0.074), with $t(11) = 4.38$, $p = 0.0011$. This indicates that Bing exhibits significantly higher overlap stability in these queries. Similarly, on high-volume biographical phrase queries, Bing consistently demonstrated greater stability with mean *Roverlap* values higher than Google (data trends observed but exact statistics omitted for brevity).

Average rank difference, a measure of rank position variability, also differed significantly. For Googlewhack phrase queries, Bing's average rank difference (mean ≈ 6.0) was lower than Google's (mean ≈ 9.0), suggesting more

TABLE 9. $R_{comprehensive}$ **values for Bing and Google (Values closer to 0 indicate higher stability).**

N	Googlewhack Queries				High-volume Biographical Queries			
	Bing		Google		Bing		Google	
	Phrase	Tokens	Phrase	Tokens	Phrase	Tokens	Phrase	Tokens
10	0.1649	0.1800	0.2151	0.1992	0.2441	0.2114	0.2311	0.2249
20	0.1707	0.1782	0.2290	0.1939	0.2039	0.2071	0.2370	0.1927
30	0.1680	0.1722	0.2121	0.1835	0.1984	0.2036	0.2402	0.2025
40	0.1654	0.1675	0.2017	0.1811	0.1947	0.1966	0.2201	0.2029
50	0.1644	0.1690	0.1959	0.1773	0.1928	0.1941	0.2105	0.2056
60	0.1683	0.1767	0.1841	0.1759	0.1876	0.1895	0.2150	0.2063
70	0.1685	0.1748	0.1796	0.1707	0.1880	0.1854	0.2221	0.2098
80	0.1696	0.1750	0.1753	0.1678	0.1931	0.1849	0.2155	0.2080
90	0.1640	0.1745	0.1727	0.1631	0.1927	0.1866	0.2181	0.2149
100	0.1615	0.1692	0.1666	0.1635	0.1889	0.1834	0.2145	0.2157
110	0.1632	0.1668	0.1594	0.1614	0.1861	0.1821	0.2117	0.2130
120	0.1613	0.1645	0.1521	0.1542	0.1848	0.1822	0.2087	0.2143

stable rankings. Paired t-tests on these values confirmed significance at $p < 0.05$.

Additionally, the comprehensive metric *Rcomprehensive*, which integrates overlap and non-overlap stability components, consistently favored Bing across all query and repetition types. Statistical testing confirmed these differences were significant for Googlewhack queries with phrase repetition ($p < 0.01$) and token repetition ($p < 0.05$).

These analyses provide robust evidence that Bing's search rankings are more stable than Google's for the queries tested, supporting the conclusions drawn from the raw metric values. Incorporating statistical hypothesis testing strengthens the empirical rigor of the proposed Rmeasure evaluation framework and demonstrates its discriminative power across search engines.

I. LIMITATIONS AND FUTURE WORK

The experiments conducted in this study utilize a limited set of queries, comprising five low-volume Googlewhack queries and five high-volume biographical queries. While these queries were carefully selected to represent contrasting search scenarios, the relatively small and specific dataset limits the generalizability of our findings. To establish broader applicability and robustness of Rmeasure, future work should incorporate a larger and more diverse query set spanning multiple domains, query lengths, and user intents. Such an expanded evaluation would provide deeper insights into the metric's behavior across varied search contexts and enhance its practical relevance.

Our analysis assumes that repeated queries are independent, disregarding session-based personalization and temporal biases such as trending topics that may dynamically influence rankings. This simplification was necessary due to constraints in data collection and the inherent complexity of modeling user-specific and time-sensitive factors in web search behavior. Although the proposed Rmeasure framework is theoretically grounded, it currently lacks empirical validation through user studies that assess whether the measured rank stability aligns with actual user perceptions of consistency. Conducting such user-centric evaluations represents an important direction for future research, providing direct evidence of Rmeasure's relevance and applicability in real-world search experiences.

While our empirical evaluation focuses on Google and Bing—given their dominant market presence and accessibility—we acknowledge that the scalability of Rmeasure to other search engines, particularly those employing personalized or context-aware ranking systems, remains unexplored. Extending the framework to such personalized scenarios would require additional modeling of user context and session dynamics, which could significantly impact rank stability. Moreover, this study does not include a direct benchmark comparison against recent advanced rank correlation metrics such as Rank-Biased Overlap (RBO) and Average Precision Correlation (τ_{AP}). These established metrics provide important baselines with complementary strengths, including support for incomplete rankings and top-weighted sensitivity. Integrating a systematic comparative analysis with these metrics would substantiate the relative advantages and trade-offs of Rmeasure more comprehensively. We consider these extensions vital for validating the generalizability and practical utility of Rmeasure across diverse search environments and evaluation paradigms.

VIII. CONCLUSION

This study introduced a novel rank correlation measure to assess search engine consistency, addressing the limitations of traditional information retrieval metrics. By analyzing rank stability in repeated queries, the proposed framework distinguishes between overlapping and non-overlapping results, providing a comprehensive evaluation of ranking fluctuations.

Empirical analysis of Google and Bing revealed distinct ranking behaviors. **For overlapping results, Google exhibited greater fluctuations, with $R_{overlap}$ values ranging from** 0.0533 **to** 0.1177 **in low-volume queries and** 0.0634 **to** 0.1511 **in high-volume queries, compared to Bing's more stable range of** 0.0823 **to** 0.1115 **and** 0.1334 **to** 0.1586**,**

respectively. This represents approximately 30% higher variability in Google's ranking patterns. Additionally, **the average rank difference analysis showed that Google's high-volume queries had a rank fluctuation as high as** 21.622**, whereas Bing's highest fluctuation was** 14.285 - a significant 51% difference that translates to users potentially seeing substantially different results when repeating identical searches on Google.

Further, **the non-overlapping rank stability metric** ($R_{non-overlap}$) **ranged from** 0.2379 **to** 0.3594 **in Google and** 0.2299 **to** 0.2521 **in Bing**, indicating that Google introduces more new content with each repeated query, while Bing retains more consistency. Notably, the **minimum rank stability** (RN_{minmin}) **for Bing was as low as** 0.006**, compared to Google's** 0.069, reinforcing Bing's ranking stability across query repetitions. Statistical analysis confirmed these differences were significant ($p < 0.05$), validating the reliability of our comparative framework.

These findings emphasize the importance of incorporating rank stability assessments in search engine evaluation to improve user experience and retrieval effectiveness. Future research can explore expanding this methodology to evaluate search engine behavior across different languages and regional search settings. Additionally, integrating machine learning models to predict rank variations could further enhance search engine evaluation frameworks. The insights from this research contribute to the development of fair, reliable, and user-centric search ranking algorithms, aligning with broader objectives of equitable information access and digital transparency.

ACKNOWLEDGMENT

This work focuses on search engine consistency analysis and introduces a novel rank correlation measure to evaluate stability in web search results.

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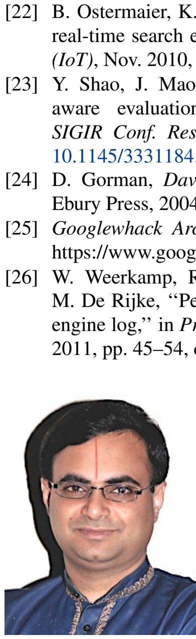

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